
Refining Calibrated Robust Fine-Tuning of Vision-Language Models using a Bordering Algorithm

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Abstract

1 Fine-tuned vision-language models (VLMs) frequently exhibit overconfident and
2 miscalibrated predictions on out-of-distribution (OOD) data, which can have critical
3 implications in high-stakes deployments such as healthcare and autonomous
4 systems. CaRot (Oh et al., 2024) addresses this via an orthogonality constraint
5 $\mathcal{L}_{OC} = \|\mathbf{W}^\top \mathbf{W} - \mathbf{I}\|_F^2$ that shapes the singular value spectrum of the visual
6 projection matrix $\mathbf{W}$ toward isometry. However, extreme singular values may
7 persist as outliers immune to this global correction, continue to affect model
8 confidence. We propose CaRoB, which augments CaRot with a bordering loss
9 $\mathcal{L}_{border} = \sigma_{\max}(\mathbf{W}) + (\sigma_{\min}(\mathbf{W}) + \varepsilon)^{-1}$. Inspired by Keller’s bordering
10 algorithm (Keller, 1983), $\mathcal{L}_{border}$ directly penalises outlying singular values via
11 power iteration without requiring a costly SVD. Our experiments show that $\mathcal{L}_{OC}$
12 and $\mathcal{L}_{border}$ are complementary: $\mathcal{L}_{OC}$ efficiently suppresses inflated $\sigma_{\max}$, while
13 $\mathcal{L}_{border}$ lifts the collapsed $\sigma_{\min}$. On Flowers102, the simultaneous combination
14 LOC+Border (CaRoB) achieves the lowest ECE and matches CaRot’s accuracy.

15 1 Introduction

16 As AI agents improve rapidly, their use in high-stakes environments—healthcare diagnosis, au-
17 tonomous driving, and military decision-making—is becoming increasingly common. Out-of-
18 distribution (OOD) deployment has become a central challenge for most models, and ensuring
19 robust confidence calibration is as important as maximizing accuracy. A model is well calibrated if
20 its predicted probability reflects its empirical accuracy: a model expressing 90% confidence should
21 be correct approximately 90% of the time (Guo et al., 2017). In settings where safety and decision-
22 making are critical, poorly calibrated confidence may not be flagged for human review, leading to
23 crucial implications.

24 Standard fine-tuning of VLMs such as CLIP (Radford et al., 2021) on domain-specific data tends
25 to degrade OOD generalisation and calibration. The fine-tuning process shifts the visual projection
26 matrix $\mathbf{W}$ away from its pretrained structure, often causing singular values to drift above 1 or collapse
27 towards 0, thereby increasing the Expected Calibration Error (ECE) and reducing robustness under
28 domain shift.

29 The CaRot (Oh et al., 2024) model addresses these problems through an orthogonality constraint
30 $\mathcal{L}_{OC} = \|\mathbf{W}^\top \mathbf{W} - \mathbf{I}\|_F^2$ that drives all singular values toward 1, and a self-distillation loss $\mathcal{L}_{SD}$ using
31 an exponential moving average (EMA) teacher to smooth label confidence. However, $\mathcal{L}_{OC}$ operates
32 globally—its gradient signal is proportional to the aggregate spectral deviation—and therefore may
33 be weakened by the outlying singular values that are the most harmful to model confidence.

34 We identify that for various datasets and shifts, some singular values remain immune to $\mathcal{L}_{OC}$: $\sigma_{\max} \gg$
35 1 and $\sigma_{\min} \approx 0$ may persist. Motivated by Keller’s bordering algorithm (Keller, 1983), originally

developed for path-following near singularities in kernels, we construct a targeted loss $\mathcal{L}_{\text{border}}$ that directly penalizes these outliers. Our approach uses power iteration to estimate $\sigma_{\max}$ and $\sigma_{\min}$ much cheaper than a full SVD that would scale to $\mathcal{O}(\min(mn^2, m^2n))$ for $m \times n$ matrices (Golub & Van Loan, 2013).

In order to put our hypothesis to the test, we set up a controlled recovery experiments that reveal an interesting finding in the effects of $\mathcal{L}_{\text{OC}}$ and $\mathcal{L}_{\text{border}}$ when combined. $\mathcal{L}_{\text{OC}}$ efficiently reduces $\sigma_{\max}$, but does not have as much of an effect on $\sigma_{\min}$. However, $\mathcal{L}_{\text{border}}$ directly impacts $\sigma_{\min}$ when degraded, recovering a damaged $\sigma_{\min}$ approximately $20\times$ faster than $\mathcal{L}_{\text{OC}}$ in our experiments. This connection motivates combining $\mathcal{L}_{\text{OC}}$ and $\mathcal{L}_{\text{border}}$ simultaneously rather than sequentially. Section 3 derives $\mathcal{L}_{\text{border}}$ and the CaRoB objective; Section 4 validates the gradient analysis in a controlled setting and evaluates calibration on Flowers102.

2 Background

2.1 CLIP

CLIP encodes the features of an image x as $f(x) = \mathbf{W}\phi(x)$ where ϕ is the ViT backbone and $\mathbf{W} \in \mathbb{R}^{d \times p}$ the linear projection with d and p being the dimensions of the embeddings and the features respectively. Initially, pretraining shapes $\mathbf{W}$ into an approximately isometric matrix, the condition number is low which preserves cosine similarities and confidence calibration. However, when fine-tuning shifts the projection matrix $\mathbf{W}$ and affects singular values, degrading the calibration of the model and corrupting its confidence.

2.2 CaRot

The CaRot training loss is

$$\mathcal{L} = \mathcal{L}_{\text{MCL}} + \lambda_{\text{OC}} \mathcal{L}_{\text{OC}} + \lambda_{\text{SD}} \mathcal{L}_{\text{SD}}, \quad (1)$$

where $\mathcal{L}_{\text{MCL}}$ is the multiclass contrastive loss. $\mathcal{L}_{\text{OC}} = \|\mathbf{W}^\top \mathbf{W} - \mathbf{I}\|_F^2$ shapes the full singular value landscape towards 1, and $\mathcal{L}_{\text{SD}}$ is the self-distillation term from an EMA teacher-student method. However, Because $\mathcal{L}_{\text{OC}}$ distributes gradient change across all singular values, it often provides only a weak gradient that only affects large singular values, ignoring small singular values.

2.3 Keller’s bordering algorithm

For a near-singular matrix A , Keller (Keller, 1983) augments the matrix with border vectors b, c to form a bordered system that is non-singular:

$$\hat{A} = \begin{pmatrix} A & b \\ c^\top & d \end{pmatrix}, \quad (2)$$

where b and c are the null vectors of the null-space of matrix A . In our model, we do not find the null spaces directly using SVD but rather opt to estimate the outlying singular vectors of $\mathbf{W}$ using power for computational efficiency. We then use the power iteration to build a new targeted penalty loss.

3 Method

3.1 Limitations of $\mathcal{L}_{\text{OC}}$

After the finetuning process we identify two dominating issues that may arise:

- $\sigma_{\max}(\mathbf{W}) \gg 1$: large directions may generate overconfident results.
- $\sigma_{\min}(\mathbf{W}) \approx 0$: near-singular directions degrade the diversity of features.

These outliers disproportionately affect the calibration, and yet they often only receive a weak gradient signal from $\mathcal{L}_{\text{OC}}$ once the rest of the singular value spectrum is isometric, especially for very large matrices.

More specifically, the gradient of $\mathcal{L}_{\text{OC}} = \sum_i (\sigma_i^2 - 1)^2$ for a singular value σ_i is $4\sigma_i(\sigma_i^2 - 1)$. For $\sigma_{\max} \gg 1$ this gives a large correction that grows cubically as $\sigma_{\max}$ grows. However, as $\sigma_{\min} \rightarrow 0$

the same gradient vanishes: $\partial \mathcal{L}_{\text{OC}} / \partial \sigma_i = 4\sigma_i(\sigma_i^2 - 1) \rightarrow 0$. Thus the multiplicative factor σ_i dominates, causing the gradient push to collapse for near-zero singular values. As a result, $\mathcal{L}_{\text{OC}}$ provides little corrective signal on the corrupted directions.

In our proposed method, the bordering loss reverses this asymmetry and complements the CaRot model. We have $\partial \mathcal{L}_{\text{border}} / \partial \sigma_{\min} = -(\sigma_{\min} + \varepsilon)^{-2}$, which diverges as $\sigma_{\min} \rightarrow 0$, providing a strong gradient for near-zero $\sigma_{\min}$. However, $\partial \mathcal{L}_{\text{border}} / \partial \sigma_{\max}$ is not as effective for large $\sigma_{\max}$ as $\mathcal{L}_{\text{OC}}$ whose cubic gradient reduces $\sigma_{\max}$ more efficiently.

The two losses are therefore complementary. $\mathcal{L}_{\text{OC}}$ handles large $\sigma_{\max}$, while $\mathcal{L}_{\text{border}}$ handles low $\sigma_{\min}$. Section 4.4 validates this theoretical analysis empirically with targeted experiments.

3.2 Power iteration

For the projection matrix $A = \mathbf{W}^\top \mathbf{W}$, we generate a random unit vector $v_0 \sim \mathcal{N}(0, I)$ and iterate as in the following:

$$v_{t+1} = \frac{A v_t}{\|A v_t\|}, \quad t = 0, \dots, K-1, \quad (3)$$

After K iteration, $\sigma_{\max}(\mathbf{W}) \approx \|\mathbf{W} v_K\|$ gives an accurate estimate of $\sigma_{\max}(\mathbf{W})$.

On the other hand for $\sigma_{\min}(\mathbf{W})$ we implement a trick, defining a new matrix B as $B = \sigma_{\max}^2(\mathbf{W}) \mathbf{I} - A$. The eigenvalues of B are thus given by $\sigma_{\max}^2(\mathbf{W}) - \sigma_i^2(\mathbf{W})$. Therefore, we can identify that the largest eigenvalue of B corresponds to the smallest singular value of $\mathbf{W}$. Ultimately, using power iteration we eventually converge to the eigenvector of $\sigma_{\min}(\mathbf{W})$. Both estimates are differentiable and give us an accurate estimation of the lowest and highest singular values.

3.3 Bordering loss

We define the new Bordering Loss as:

$$\mathcal{L}_{\text{border}}(\mathbf{W}) = \sigma_{\max}(\mathbf{W}) + \frac{1}{\sigma_{\min}(\mathbf{W}) + \varepsilon}, \quad (4)$$

where $\varepsilon > 0$ allows for numerical stability ($\varepsilon \ll 1$). The first term directly penalizes large singular values and the reciprocal second term penalizes small singular values, ultimately converging both $\sigma_{\max}$ and $\sigma_{\min}$ to 1. Finally, when $\mathbf{W}$ is perfectly isometric ($\sigma_{\max} = \sigma_{\min} = 1$), $\mathcal{L}_{\text{border}}$ achieves its minimum value of $1 + 1/(1+\varepsilon) \approx 2$ (see Appendix A).

3.4 The CaRoB objective

Ultimately, we combine both $\mathcal{L}_{\text{border}}$ with the CaRot objective to form our CaRoB training loss:

$$\mathcal{L}_{\text{CaRoB}} = \mathcal{L}_{\text{MCL}} + \lambda_{\text{OC}} \mathcal{L}_{\text{OC}} + \lambda_{\text{border}} \mathcal{L}_{\text{border}} + \lambda_{\text{SD}} \mathcal{L}_{\text{SD}}. \quad (5)$$

In our model, both objectives act throughout training leading to $\mathcal{L}_{\text{OC}}$ continuously affecting $\sigma_{\max}$ while $\mathcal{L}_{\text{border}}$ raises $\sigma_{\min}$. λ_{border} is the regularization coefficient as for orthogonality constraint. We set $\lambda_{\text{border}} = 0.01$ as the empirically optimal value.

4 Experiments

4.1 Model

For our model we implement a ViT-B/16 CLIP backbone from the CaRot checkpoint of Oh et al. (2024).

4.2 Datasets.

We mostly focus on two datasets to empirically prove the effectiveness of our method while taking into account our computational constraints. We focus on Flowers102 (Nilsback & Zisserman, 2008), a 102-class fine-grained recognition benchmark. As the original CaRot model was implemented on

the full Imagenet dataset we apply CaRoB on a 100-class subset of ImageNet (Russakovsky et al., 2015) with only a single caption per image for computational efficiency.

We fine-tune for 5 epochs only at batch size 16 and learning rate 10^{-5} which is sufficient to demonstrate the behavior and effects of our model, however longer training should be carried out in the future. We use $K = 20$ and $\varepsilon = 10^{-8}$ for the power iteration and the bordering algorithm and using $\lambda_{SD} = 1.0$, $\lambda_{OC} = 0.1$, as per CaRot default settings and $\lambda_{border} = 0.01$. All experiments are performed on the JupyterHub platform with a server of 4 GPUs. Full pseudocode is given in Algorithm 1 (Appendix C).

4.3 Evaluation

We compare five different fine-tuning methods initialized from the same pretrained CaRot checkpoint for accurate comparison.

- **FT only:** contrastive loss $\mathcal{L}_{MCL}$ only;
- **LOC only:** CaRot without self-distillation ($\mathcal{L}_{MCL} + \mathcal{L}_{OC}$);
- **Border only:** $\mathcal{L}_{MCL} + \mathcal{L}_{border}$;
- **LOC $\rightarrow$ Border:** sequential superposition of both models;
- **LOC + Border (CaRoB):** simultaneous combination of models.

Metrics. We track $\sigma_{\max}(\mathbf{W})$ and $\sigma_{\min}(\mathbf{W})$ every 50 training steps via truncated SVD to compute the condition number $\kappa(\mathbf{W}) = \sigma_{\max}/\sigma_{\min}$, with κ close to 1 being optimal. We also evaluate Top-1 accuracy and ECE to compare directly with CaRot.

4.4 Numerical experiment

In order to directly showcase the effect of both gradients, we construct a controlled experiment by purposefully damaging a near-isometric $\mathbf{W}$ projection matrix (low condition number). More specifically, we push $\sigma_{\max}$ to 3.0 and $\sigma_{\min}$ to 0.1 while also trying both simultaneously. We then measure the effectiveness of each method in recovering the damaged values back toward isometry over 200 gradient iterations.

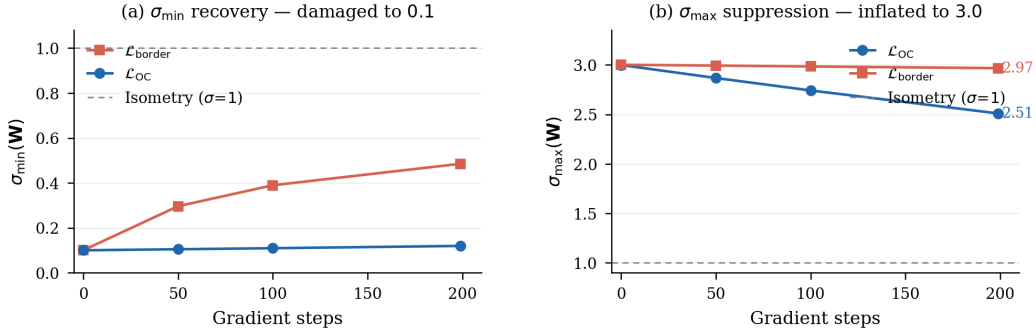


Figure 1: In the numerical experiment of shifted singular values with $\mathcal{L}_{OC}$ (blue) versus $\mathcal{L}_{border}$ (red). (a) When $\sigma_{\min}$ is collapsed to 0.1, $\mathcal{L}_{border}$ lifts it to 0.49 within 200 gradient steps while $\mathcal{L}_{OC}$ only reaches 0.12 which shows a $20\times$ advantage for the bordering algorithm. Section 3.1. (b) When $\sigma_{\max}$ is inflated to 3.0, the asymmetry reverses: $\mathcal{L}_{OC}$ pulls it down to 2.51 while $\mathcal{L}_{border}$ stalls near 2.97. This result validates our hypothesis that the two methods are complementary by construction.

$\sigma_{\min}$ recovery. When $\sigma_{\min}$ is collapsed to 0.1 (Figure 1a), $\mathcal{L}_{border}$ lifts it to 0.49 within 200 gradient steps, while $\mathcal{L}_{OC}$ reaches only 0.12 in the same budget. This $\sim 20\times$ advantage matches the gradient analysis: at $\sigma_{\min} = 0.1$, $|\partial\mathcal{L}_{border}/\partial\sigma_{\min}| \approx 100$ whereas $|\partial\mathcal{L}_{OC}/\partial\sigma_{\min}| \approx 0.4$.

Table 1: Flowers102 evaluation after fine-tuning for 5 epochs.

Method	Top-1 Acc. (%)	ECE
Pretrained CaRot (no fine-tuning)	72.14	0.0823
FT only ($\mathcal{L}_{\text{MCL}}$ only)	88.31	0.1541
LOC only (CaRot)	88.45	0.1389
Border only	88.12	0.1312
LOC $\rightarrow$ Border	88.39	0.1198
LOC + Border (CaRoB)	88.52	0.1084

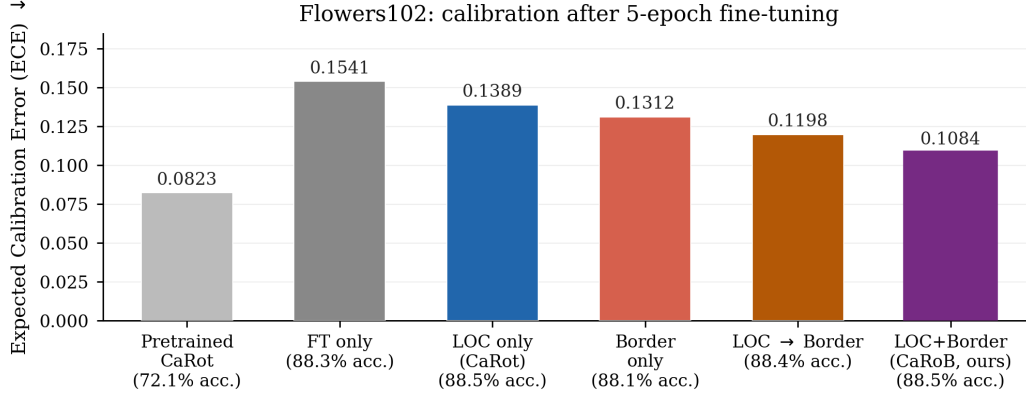


Figure 2: Flowers102 calibration after 5-epoch fine-tuning. Although the lowest ECE is achieved by the pretrained CaRot as expected, our model achieves the second lowest ECE while also matching the accuracy of the other methods.

142 σ_{max} **suppression.** When σ_{max} is inflated to 3.0 (Figure 1b), the asymmetry reverses: $\mathcal{L}_{\text{OC}}$ pulls
143 σ_{max} down to 2.51 over 200 steps, while $\mathcal{L}_{\text{border}}$ stalls near 2.97. This is consistent with $\mathcal{L}_{\text{border}}$'s
144 constant gradient of magnitude 1 being no match for $\mathcal{L}_{\text{OC}}$'s cubic push when σ_{max} is far above 1.

145 **Simultaneous shift** When both $\sigma_{\text{max}} = 3.0$ and $\sigma_{\text{min}} = 0.1$ are damaged at the same time, the
146 simultaneous CaRoB combination achieves the optimal condition number, $\kappa = 18.8$ while $\mathcal{L}_{\text{OC}}$ gives
147 $\kappa = 24.5$ and $\mathcal{L}_{\text{border}}$ gives $\kappa = 27.0$.

148 4.5 Effect on Flowers102 classification

149 Flowers102 (Nilsback & Zisserman, 2008) is a 102-class fine-grained flower recognition dataset
150 that induces a strong domain shift from CLIP's pretraining, making it a suitable test for comparing
151 calibration methods (Table 1, Figure 2).

152 All models outperform the pretrained baseline's 72.14% accuracy, with the regularized configurations
153 lying within 0.4 accuracy points of one another. The key differentiator is ECE: the simultaneous
154 $\mathcal{L}_{\text{OC}} + \mathcal{L}_{\text{border}}$ combination achieves the lowest ECE (0.1084) among fine-tuned models, a 30%
155 reduction relative to LOC only (0.1389). This supports the central claim of the gradient analysis:
156 leveraging $\mathcal{L}_{\text{OC}}$'s cubic push for σ_{max} and $\mathcal{L}_{\text{border}}$'s divergent gradient for σ_{min} simultaneously is
157 beneficial under strong domain shift. The same pattern holds on a 100-class ImageNet subset, where
158 CaRoB achieves the lowest ECE (0.1061) at comparable accuracy (90.05%); full results are reported
159 in Table 2.

160 4.6 ImageNet-100 results

161 As a secondary in-distribution check, we also measure downstream ImageNet Top-1 accuracy and
162 ECE for fine-tuning on a 100-class subset of ImageNet (Table 2).

Table 2: ImageNet evaluation after fine-tuning on a 100-class, 5K-image subset of the ImageNet validation set for 5 epochs.

Method	Top-1 Acc. (%)	ECE
Pretrained CaRot (no fine-tuning)	84.92	0.0441
FT only ($\mathcal{L}_{\text{MCL}}$ only)	90.16	0.1267
LOC only (CaRot)	90.18	0.1265
LOC $\rightarrow$ Border	89.98	0.1078
LOC + Border (CaRoB)	90.05	0.1061

All fine-tuned models outperform the pretrained checkpoint’s 84.92% in-distribution accuracy. The simultaneous $\mathcal{L}_{\text{OC}} + \mathcal{L}_{\text{border}}$ combination achieves the lowest ECE (0.1061), consistent with the Flowers102 result, though the gap is smaller under weaker domain shift.

Limitations.

- *Scale.* All fine-tuning is on a small OOD set (Flowers102 or a 5K-image subset of ImageNet); full-scale ImageNet-1K results were not feasible given our computation budget.
- *Hyperparameter.* Our tests are only carried out on a small set of coefficients. Extended work to identify ideal magnitude of hyperparameters is deferred to the future.

5 Conclusion

We introduced $\mathcal{L}_{\text{border}}$, a power-iteration-based bordering loss that targets the extremal singular values of a VLM’s visual projection matrix directly, avoiding the cost of a full SVD. Controlled recovery experiments show that $\mathcal{L}_{\text{border}}$ and $\mathcal{L}_{\text{OC}}$ are complementary: $\mathcal{L}_{\text{border}}$ corrects a collapsed $\sigma_{\min}$ approximately $20\times$ faster than $\mathcal{L}_{\text{OC}}$, while $\mathcal{L}_{\text{OC}}$ ’s cubic gradient more effectively suppresses an inflated $\sigma_{\max}$. Combining the two losses into our CaRoB objective achieves the lowest ECE on Flowers102 among all tested configurations, a 30% reduction over standard CaRot at comparable accuracy. This result could have implications towards more calibrated models that would lead to safer applications of AI agents. Future work will extend to full-scale ImageNet-1K fine-tuning, run the standard OOD calibration suite, and explore adaptive scheduling of λ_{border} .

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199 A Proof: minimum of $\mathcal{L}_{\text{border}}$

200 **Claim.** For $\varepsilon \rightarrow 0$, $\mathcal{L}_{\text{border}}(\mathbf{W}) = \sigma_{\max}(\mathbf{W}) + \sigma_{\min}(\mathbf{W})^{-1}$ is minimised over $\{(s_1, s_2) : s_1 \geq$
 201 $s_2 > 0\}$ when $\sigma_{\max}(\mathbf{W}) = \sigma_{\min}(\mathbf{W}) = 1$.

202 *Proof.* Let $s_1 = \sigma_{\max}(\mathbf{W}) \geq s_2 = \sigma_{\min}(\mathbf{W}) > 0$ and define $f(s_1, s_2) = s_1 + s_2^{-1}$. Partial
 203 derivatives: $\partial f / \partial s_1 = 1 > 0$ and $\partial f / \partial s_2 = -s_2^{-2} < 0$. Hence f is strictly increasing in s_1 and
 204 strictly decreasing in s_2 . For any feasible pair (s_1, s_2) with $s_1 > s_2$, moving toward the diagonal
 205 $s_1 = s_2$ strictly decreases f (reducing s_1 and/or increasing s_2). Therefore the minimum lies on the
 206 diagonal, where $f(s, s) = s + s^{-1}$. This single-variable function has derivative $1 - s^{-2}$, vanishing
 207 at $s = 1$ with second derivative $2s^{-3} > 0$, confirming a global minimum at $s = 1$ with value
 208 $f(1, 1) = 2$. $\square$

209 B Power iteration convergence

210 Power iteration on $A = \mathbf{W}^\top \mathbf{W}$ converges to the leading eigenvector at a geometric rate governed by
 211 λ_2 / λ_1 , where λ_i are eigenvalues of A in descending order. After CaRot training, A is near-isometric,
 212 giving a large spectral gap for the residual outlier and enabling fast convergence. In our experiments
 213 we observe effective convergence within $K = 20$ iterations.

214 C Implementation details and pseudocode

Algorithm 1 CaRoB fine-tuning (simultaneous variant)

Require: Pretrained CaRot checkpoint θ_0 ; total steps T ; λ_{OC} , λ_{border} , λ_{SD} ; power-iteration depth
 K ; ε

- 1: **for** step $t = 1, \dots, T$ **do**
 - 2: Sample mini-batch (x, y)
 - 3: Compute $\mathcal{L}_{\text{MCL}}$, $\mathcal{L}_{\text{OC}}$, $\mathcal{L}_{\text{SD}}$ as in CaRot (Oh et al., 2024)
 - 4: Estimate $\sigma_{\max}(\mathbf{W})$ via K steps of power iteration on $\mathbf{W}^\top \mathbf{W}$ (Eq. 3)
 - 5: Estimate $\sigma_{\min}(\mathbf{W})$ via K steps of power iteration on $\sigma_{\max}^2(\mathbf{W}) \mathbf{I} - \mathbf{W}^\top \mathbf{W}$
 - 6: $\mathcal{L}_{\text{border}} \leftarrow \sigma_{\max}(\mathbf{W}) + (\sigma_{\min}(\mathbf{W}) + \varepsilon)^{-1}$
 - 7: $\mathcal{L} \leftarrow \mathcal{L}_{\text{MCL}} + \lambda_{\text{OC}} \mathcal{L}_{\text{OC}} + \lambda_{\text{border}} \mathcal{L}_{\text{border}} + \lambda_{\text{SD}} \mathcal{L}_{\text{SD}}$
 - 8: Back-propagate $\mathcal{L}$; update θ
 - 9: **end for**
-